# Every word, by section and lecture

161 lines total. Shape: **page · term = also called · short meaning**. Fill the page number from your notebook. Say the meaning before you write it. One line per word; if the line's long, abbreviate the meaning, not the term.

## S1 · Satellite types & orbits

p__ · satellite · anything that orbits something bigger
p__ · orbit · falling and missing, forever
p__ · altitude · height above the ground, not from center
p__ · LEO / MEO / GEO = low / medium / geosynchronous · 200–2000 km · 2000–35,786 · 35,786
p__ · orbital period (T) · time for one lap: 90 min / 12 h / 24 h
p__ · geosynchronous (GSO) · 1-day lap, any tilt
p__ · geostationary (GEO) · 1-day lap + over equator, stays put
p__ · equator / equatorial plane · flat sheet through Earth's middle
p__ · sidereal day · true spin vs stars, 23 h 56 m
p__ · astronomical satellite = space telescope · looks out: Hubble, JWST
p__ · Earth observation (EO) · looks down: Landsat
p__ · weather satellite · looks down from GEO: GOES
p__ · reconnaissance satellite = spy sat · looks down, military
p__ · navigation satellite · broadcasts time, phone finds itself: GPS
p__ · communication satellite = comms / relay · radio up, radio down: Starlink
p__ · biosatellite · carries living things: Laika
p__ · space station · satellite people live in: ISS
p__ · constellation · swarm so one is always overhead
p__ · footprint = coverage area · patch a sat sees; bigger when higher
p__ · eclipse (orbital) · Earth blocks the Sun; batteries cover it
p__ · atomic oxygen / drag · LEO-only problems: eats plastic / slows you

## S1 · Types of interplanetary spacecraft

p__ · interplanetary · between planets, orbiting the Sun
p__ · flyby · passes once, never captured
p__ · orbiter · brakes into orbit, stays
p__ · atmospheric probe · dives into the air, short life
p__ · lander · touches down, stays put
p__ · rover · lander with wheels
p__ · penetrator = impactor · crashes in on purpose
p__ · observatory spacecraft · telescope platform, target = sky
p__ · communications spacecraft = relay · forwards messages for others
p__ · commitment ladder = mission types · flyby → orbiter → probe → lander → rover → penetrator
p__ · delta-v (Δv) · speed change = fuel bill
p__ · braking burn = capture burn · orbit insertion · slow down so gravity grabs you
p__ · prograde / retrograde · burn to speed up / slow down
p__ · gravity assist = flyby · slingshot · steal speed from a moving planet
p__ · EDL = entry, descent, landing · 7 minutes of terror
p__ · C3 = characteristic / launch energy · how hard you throw it from Earth
p__ · Kuiper Belt · icy ring beyond Neptune, Pluto lives there
p__ · comet vs asteroid · ice + dust with a tail / a rock

## S2 · Space vacuum conditions

p__ · vacuum · almost no gas: no pressure, no air
p__ · ozone layer ($O_3$) · Earth's sunscreen, blocks UV < 0.3 µm
p__ · ultraviolet (UV) · invisible high-energy light, sunburns

p__ · wavelength · size of one ripple; shorter = more energy
p__ · µm (micrometer) = micron · millionth of a meter; 0.3 µm = 300 nm
p__ · polymer = plastic · long molecule chains: Kapton, Teflon
p__ · organic chemical bond = bond energy · carbon glue; UV photon beats it
p__ · degradation / decomposition · material falling apart over time
p__ · corrosion · chemical attack (rust); loose word for UV damage
p__ · UV photodegradation · photons snap polymer bonds
p__ · outgassing · gas leaking out of solids in vacuum
p__ · TML / CVCM · outgassing limits: < 1% / < 0.1%
p__ · bake-out · outgas it on the ground first
p__ · void · bubble inside a material
p__ · pressure difference · 1 atm inside, 0 outside → cracks
p__ · clean room · filtered air, no specks on sensors
p__ · perturbation · small unwanted nudge
p__ · dissipation = damping · energy leaking away until motion stops
p__ · Newton's 1st law = inertia · motion keeps going unless pushed
p__ · vibration / damping · shaking / what makes it fade
p__ · convection · heat carried by moving air; gone in vacuum
p__ · conduction · heat through touching solids
p__ · radiation (heat) · heat glowing away; works in vacuum
p__ · radiator · panel that glows heat into space
p__ · MLI = multilayer insulation · thermal blanket · crinkly gold wrap
p__ · equilibrium / hot spot · settled temperature / unspread heat
p__ · electrical discharge = ESD · a spark
p__ · spacecraft charging · static builds, no air to bleed it, arcs

# Every word, by section and lecture

161 lines total. Shape: **page · term = also called · short meaning**. Fill the page number from your notebook. Say the meaning before you write it. One line per word; if the line's long, abbreviate the meaning, not the term.

p__ · passive vs active system · no power (coating) / powered (heater)

## S2 · Earth atmosphere

p__ · atmosphere · blanket of gas held by gravity
p__ · troposphere · 0–11 km, weather, colder up
p__ · stratosphere · 11–50 km, ozone, warmer up
p__ · mesosphere · 50–90 km, coldest, meteors burn
p__ · thermosphere · 90–500 km, hot but empty, ISS here
p__ · exosphere · 500 km+, fades to space
p__ · homosphere / heterosphere · stirred mix / sorted by weight
p__ · turbopause = homopause · ~100 km, mixing stops
p__ · temperature vs heat · molecule speed / energy delivered
p__ · density / number density · how much stuff per volume ($m^{-3}$)
p__ · log scale · each tick = 100,000×
p__ · constituent = component · one gas in the mix
p__ · $N_2$ / $O_2$ / Ar / $CO_2$ · 78% / 21% / 0.9% / 0.04%
p__ · O / N / H / He = atomic forms · single atoms, up high
p__ · atomic oxygen (AO) · lone O atom, 200–600 km, eats plastic
p__ · photodissociation · light splits $O_2 \rightarrow O + O$
p__ · anomalous O = hot oxygen · minor extra-hot O above 300 km
p__ · ionosphere · 60–1000 km, charged, bends radio, delays GPS
p__ · geocorona · hydrogen halo glowing in UV
p__ · VLEO / LEO · below ~250 km heavy drag / up to 2000
p__ · reactive · eager to bond, sticky
p__ · oxidation · oxygen bonding to a material; rust
p__ · erosion · worn away bit by bit
p__ · erosion yield / fluence · loss per atom / total atoms that hit

p__ · kinetic energy / relative velocity · energy of motion / closing speed
p__ · eV (electron-volt) · tiny energy unit; bond ≈ few eV
p__ · ram / wake · front face / sheltered back
p__ · atmospheric drag · thin air slows LEO sats; ISS reboosts
p__ · Kapton = polyimide film · gold insulation, AO's victim
p__ · F10.7 / Ap = space weather indices · Sun activity; puffs up the air

## S3 · Orbit definition & parameters

p__ · orbital mechanics · math of where a s/c goes under gravity
p__ · celestial body · planet, moon, Sun, comet
p__ · transfer orbit · the highway between two orbits
p__ · Classical Orbital Elements = Keplerian elements · the six · a e i Ω ω ν = the hoop's settings
p__ · semi-major axis (a) · size: $(r_p + r_a)/2$
p__ · eccentricity (e) · squash: 0 circle → 1 cigar → >1 open
p__ · perigee / apogee = periapsis / apoapsis · closest, fastest / farthest, slowest
p__ · perihelion / aphelion · same, around the Sun
p__ · apsis (apsides) · either extreme point
p__ · line of apsides = major axis · perigee↔apogee line
p__ · inclination (i) · tilt vs equator; 0° flat, 90° polar
p__ · ecliptic · plane of Earth's orbit; reference for Sun orbits
p__ · plane of reference · what you measure tilt from
p__ · vernal equinox (♈) = First Point of Aries · reference direction · fixed zero direction in space
p__ · ascending node · hoop crosses equator going north

p__ · line of nodes · where orbital plane cuts equator
p__ · RAAN (Ω) = longitude of ascending node · swivel: ♈ → ascending node
p__ · argument of periapsis (ω) = argument of perigee · spin in plane: node → perigee
p__ · true anomaly (ν) · where the bead is now
p__ · epoch · timestamp the elements are valid for
p__ · orbital plane · flat sheet the orbit lies in; passes through center
p__ · circular / elliptical / hyperbolic · e = 0 / 0–1 / > 1
p__ · polar / equatorial orbit · i = 90° / i = 0°
p__ · geometry (of an orbit) · size + shape + orientation
p__ · vis-viva · $v^2 = \mu(2/r - 1/a)$
p__ · μ (gravitational parameter) = GM · Earth 398,600 $km^3/s^2$

## S3 · Earth orbits & constellations

p__ · Earth orbit = geocentric · goes around Earth, not the Sun
p__ · rotation period · Earth spins once: 23 h 56 m
p__ · altitude vs distance from center · GEO 35,786 alt = 42,164 from center
p__ · coverage / global coverage · what a sat sees / everything, always
p__ · resolution · how much detail; closer = sharper
p__ · orbital plane (constellation) = Walker constellation · several rings so sats aren't bunched
p__ · revisit time · how long until same spot again
p__ · GPS / GLONASS / Galileo / BeiDou · US / Russia / EU / China nav constellations
p__ · Iridium · 66 sats, 780 km, sat phones anywhere

# Every word, by section and lecture

161 lines total. Shape: **page · term = also called · short meaning**. Fill the page number from your notebook. Say the meaning before you write it. One line per word; if the line's long, abbreviate the meaning, not the term.

p__ · data relay / transfer time · sat → ground; higher = more delay
p__ · trade-off = compromise · better coverage costs detail + delay
p__ · sun-synchronous (SSO) · polar-ish LEO, same local sun time
p__ · $J_2$ / nodal precession · Earth's bulge swivels Ω; makes SSO work

## S3 · Transfers & flybys

p__ · Lambert's problem · 2 points + a time → the orbit + Δv
p__ · input / constraint · what you feed in / limit you can't break
p__ · conic / patched conics = conic approximation · stitched 2-body pieces of a trajectory
p__ · sphere of influence (SOI) · where a planet's gravity wins; Earth ≈ 925,000 km
p__ · Hohmann transfer · lazy half-ellipse, 180°, 2 burns, cheapest
p__ · tangent · just touching; why burns are small
p__ · optimal / efficient · least Δv, not fastest
p__ · time of flight (TOF) · Hohmann = half the ellipse's period
p__ · launch window / synodic period · lineup repeats; Mars every 26 months
p__ · phase angle = lead angle · how far ahead target must be: Mars ~44°
p__ · impulsive burn · engine firing treated as instant
p__ · circularization burn = apogee kick · 2nd burn rounds the ellipse
p__ · plane change · tilting the orbit; very expensive
p__ · bi-elliptic transfer · 3-burn alternative, only for huge jumps
p__ · gravity field · region where the planet's pull matters
p__ · relative velocity · speed as seen from Mars vs from Sun
p__ · vector / magnitude / direction · arrow / its length / which way

p__ · $v_\infty$ (hyperbolic excess velocity) · speed relative to planet, far away
p__ · turning angle (δ) = deflection angle · how much the flyby bends you
p__ · orbital energy = specific orbital energy · speed + height budget; bigger = bigger a
p__ · aerobraking · use atmosphere to shed speed instead of fuel
p__ · Tsiolkovsky rocket equation = ideal rocket equation · $\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$
p__ · Rosetta / 67P / ESA · comet mission 2004→2014, 4 flybys
p__ · deep space · beyond the Moon; outer planets